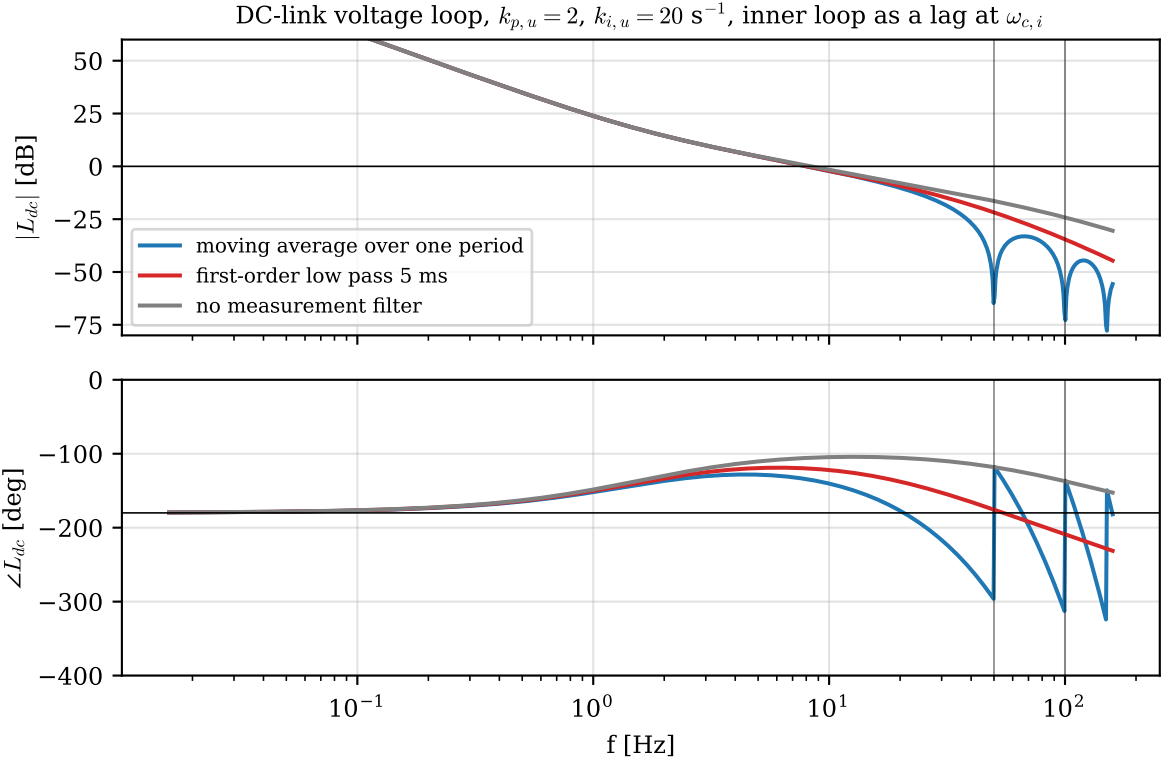
DC-link voltage loop, $k_{p,u} = 2$, $k_{i,u} = 20$ s^{-1}, inner loop as a lag at $\omega_{c,i}$
$|L_{dc}|$ [dB]
50
25
0
−25
−50
−75
moving average over one period
first-order low pass 5 ms
no measurement filter
$\angle L_{dc}$ [deg]
0
−100
−200
−300
−400
10^{-1}
10^{0}
10^{1}
10^{2}
f [Hz]